Engs 28

Embedded Systems

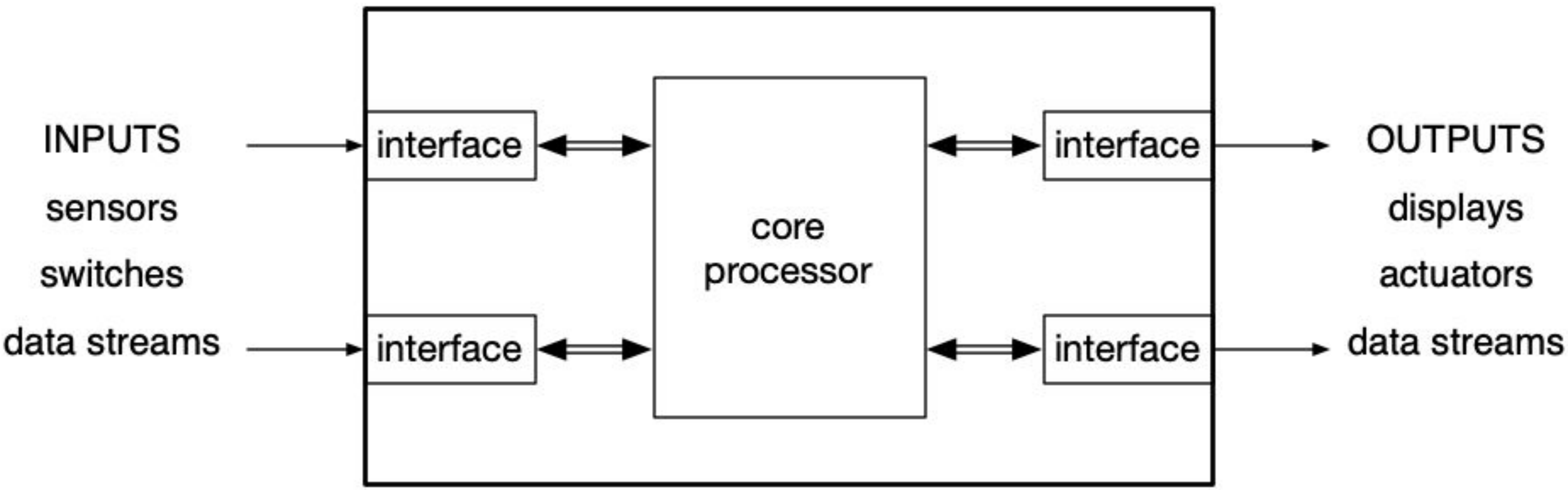
Day 5x

Pick a name for your table group!

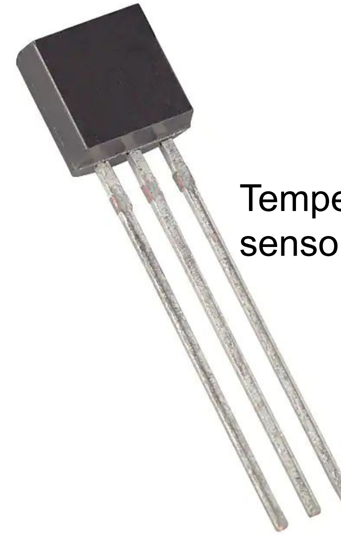
Brainstorm a team name
Let your LF know your group name

Inputs and Outputs to an Embedded System

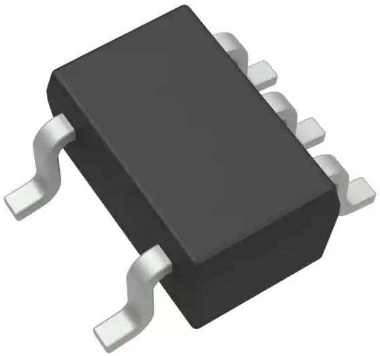
Top-level embedded system view



Input Component Examples

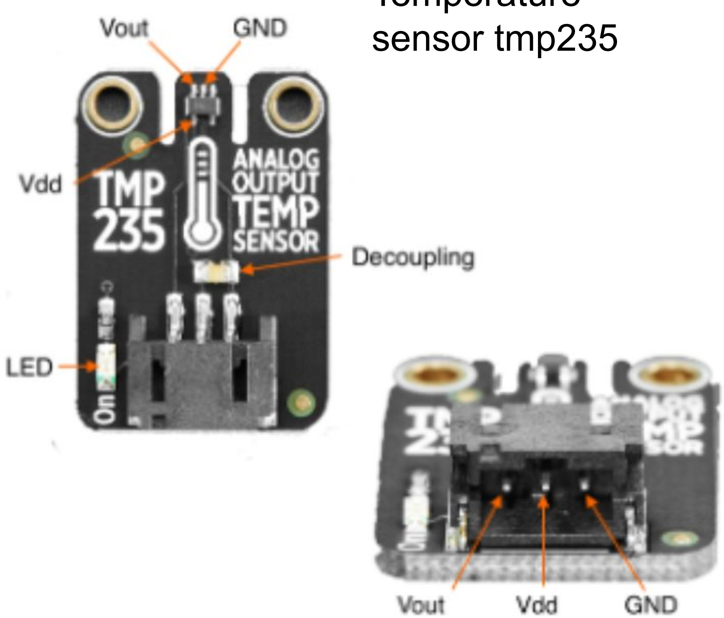


Temperature sensor tmp36



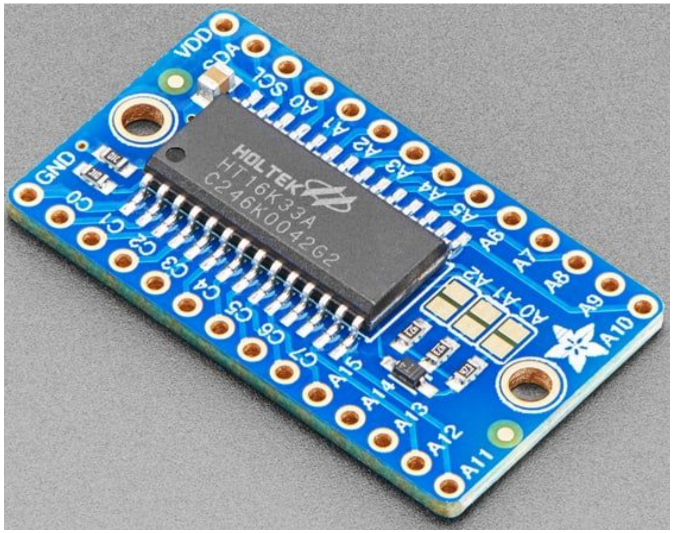
CdS photocell

Temperature sensor tmp235

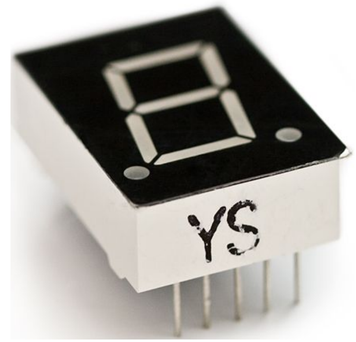


Accelerometer LSM303AGR

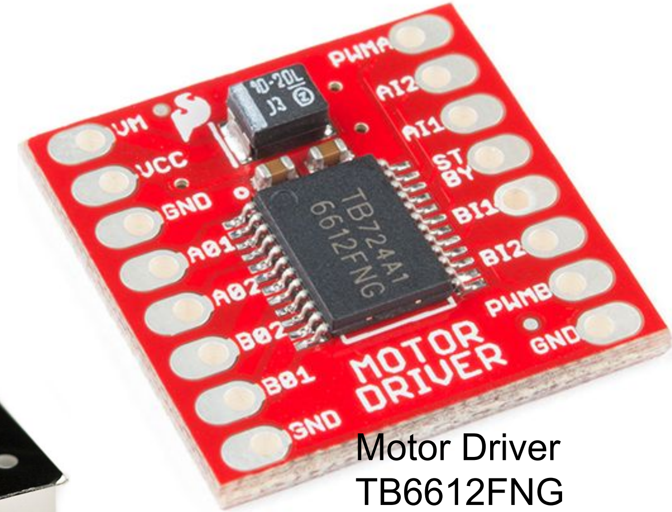
Output Component Examples



LED Matrix Driver
Backpack HT16K33



7-segment display
YSD-160AR4B-8



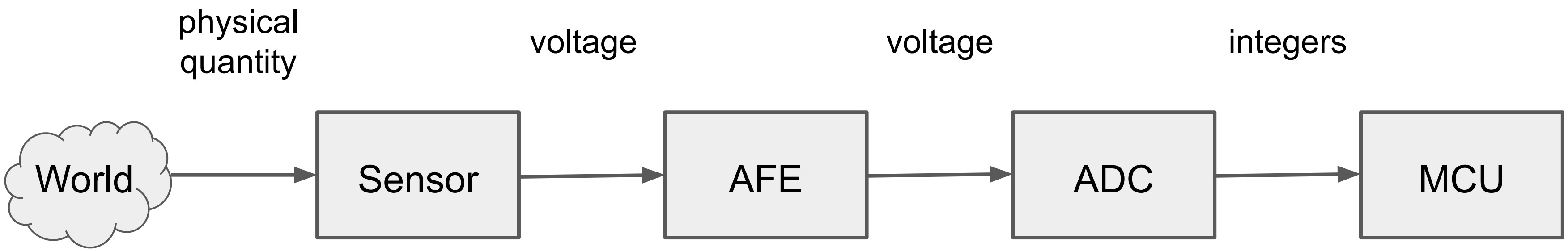
Motor Driver
TB6612FNG

Servomotor

LED
L513SRD-C

DC motor

Typical sensor signal chain



Sensor = Mechanism that translates one physical variable into another

AFE = Analog front end: Amplify, reduce noise (a.k.a. signal conditioning)

ADC = Analog to digital converter

MCU = Microcontroller unit (i.e., our STM32C031C6)

Many sensors are “bare” (example: thermocouple, phototransistor, force-sensitive resistor)

Often, the sensor you buy has the AFE built in (example: TMP235 temperature sensor)

Often, the sensor you buy has the AFE and the ADC built in, and a serial communication interface. (example: LSM303AGR accelerometer / magnetometer)

Usually, the MCU has an ADC built in (ours does).

Interfacing with Input and Output Components
Aka Reading a Data Sheet

Note: Microcontrollers, such as our STM32C031C6, also have datasheets. That's where we found the registers for the USART module!

Here, we'll focus on sensors and actuators.

Datasheets

- Datasheets are the roadmap to a sensor or an actuator (or a chip for that matter).
- Every circuit part has a data sheet (even a resistor!)
- Datasheets are packed densely with information.
- Datasheets can be intimidating.
- Reading datasheets requires patience.
- Datasheets vary between component types and manufacturers.

"In some ways, reading a datasheet is like coming into the middle of a technical conversation." (Elicia White, Embedded Software Engineer)

Datasheet HiTA Activity

As a table group, please complete the "Learn to read datasheets" activity in HiTA. HiTA will pick one of these datasheets for your activity:

- Accelerometer
- Temperature sensor
- Real Time Clock
- Hall Effect
- Humidity sensor

After HiTA tells you which datasheet you'll be working with, open it up and display it on your projector. There are links to all datasheets on our Canvas front page.

Summary - Datasheets: The Hardware Side

First page typically shows at least:

- Features
- Description - ALWAYS READ THIS

Most parts then have:

- Pinout
- Electrical characteristics
- Absolute maximum ratings
- Packaging information
- Recommended operating conditions

If a pin has a star next to it or a line over the name, it means that the pin is active low.
You'll pull the pin low (0V) to activate it, rather than high (VCC)

Some parts follow with performance characteristics, timing diagrams, example schematics.

At the end of many datasheets is packaging information (dimensions of the packages a part is available in).

Datasheets: Interfacing With The Component

Look for a section "Application Information" or "Theory of Operation" or "Detailed Description" or some such thing about halfway through the datasheet.

- Read this section carefully.
- Mark areas that may be relevant for your code.
  - Analog sensors will output a voltage that you'll need to interpret.
  - Digital sensors will communicate, via a protocol, some information that you'll need to read and interpret.
  - Analog actuators need to be sent a specific voltage in order to perform a specific task.
  - Digital actuators need to receive information, via a protocol, with instructions.
- Pay particular attention to:
  - How to connect your component to a microcontroller
  - How to read its output, or send signals to it
  - How to calibrate it, etc.

Create one slide for your component

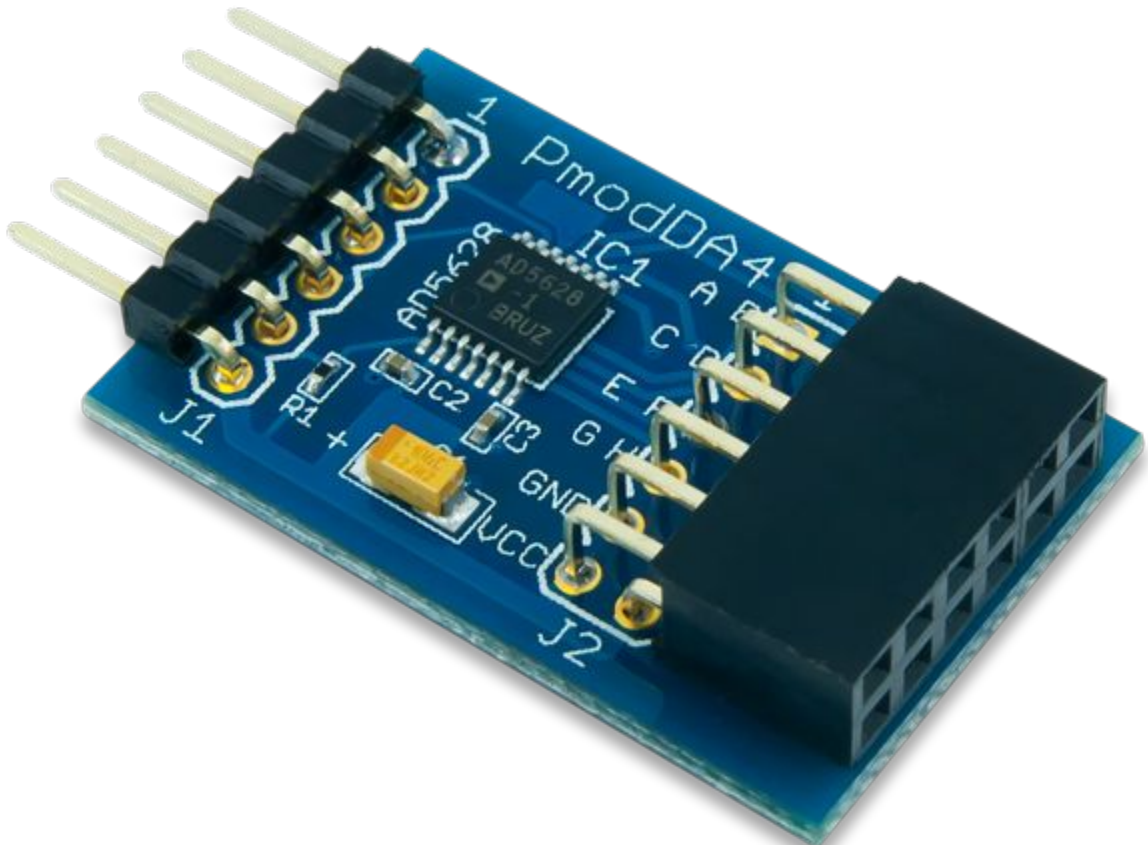
Include on your slide:

- Your component's name, model number and manufacturer
- A brief description of your component
- How to connect your component to a microcontroller
- How to communicate with your component
- Anything else that seems crucial to include

Put your group's slide here.

Find a Datasheet

Find the datasheet for the device your table is given.



What is the absolute maximum power supply voltage (VDD or VCC) that can be applied to the part?